

Sri Lanka Institute of Information Technology



BSc (Hons) in Information Technology

Specializing in Data Science

IT3021 – Data Warehousing and Business Intelligence

Year 3

Assignment 2

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1. Introduction

This assignment presents the implementation of a comprehensive Business Intelligence (BI) solution using Microsoft SQL Server technologies. This is an integrated solution comprising of data warehouse, SSAS cube, OLAP analysis in Excel and Power BI dashboards.

Databases where the primary goal is to analyze transactional data using multidimensional models and perform OLAP operations like roll-up, drill-down, slice, dice, pivot. Interactive reports are also created using Excel and Power BIs, which aids in proper decision-making.

2. DATA SOURCE DESCRIPTION

The data source for this assignment is the data warehouse developed in Assignment. The warehouse follows a star schema design consisting of one fact table and multiple dimension tables.

Fact Table:

- FactTransaction

Dimension Tables:

- DimCustomer
- DimAccount
- DimCard
- DimLoan
- DimDate

Description

The fact table stores transactional measures such as transaction amount, transaction count, account balance after transactions, and transaction processing time. The dimension tables provide descriptive attributes related to customers, accounts, cards, loans, and time.

This structure enables efficient analytical queries and supports multidimensional analysis.

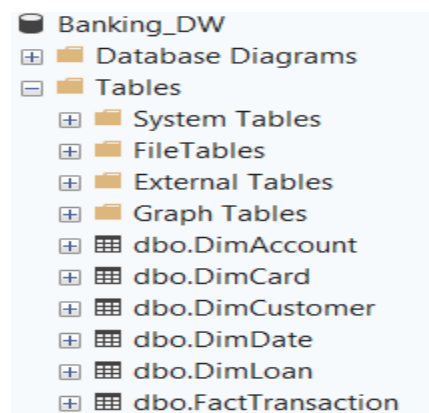


Figure 1: Data warehouse tables

SSAS CUBE IMPLEMENTATION

An SSAS cube was created using the data warehouse as the data source to enable multidimensional analysis.

Steps Followed

1. Created a new Analysis Services Project
2. Configured Data Source connection
3. Created Data Source View (DSV)
4. Selected FactTransaction as the fact table
5. Added related dimension tables
6. Defined measures and dimensions
7. Deployed the cube successfully

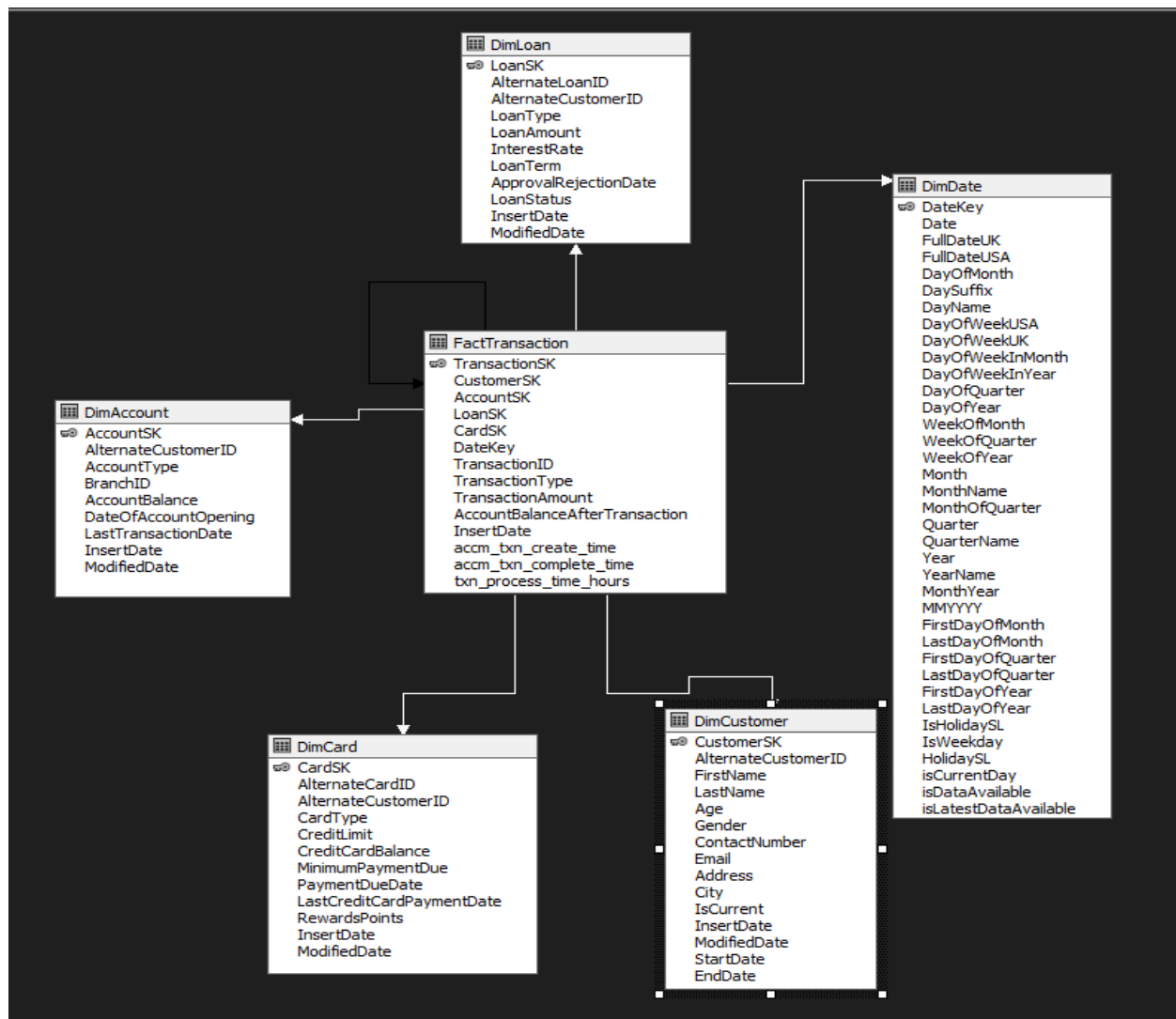


Figure 2: Data Source View

Measures Used

- Transaction Amount
- Fact Transaction Count
- Account Balance After Transaction
- Txn Process Time Hours

Dimensions Used

- Customer Dimension
- Account Dimension
- Card Dimension
- Loan Dimension
- Date Dimension

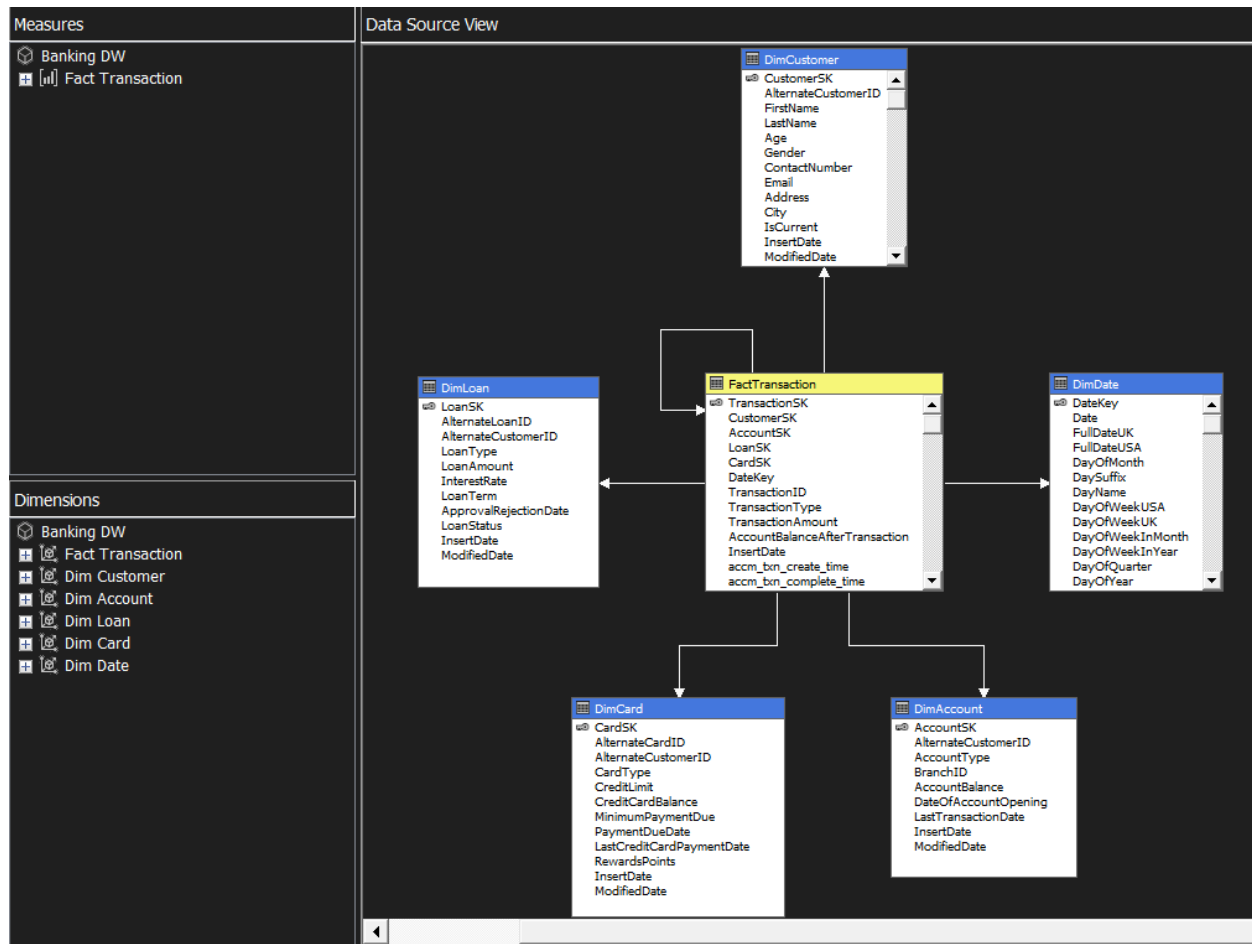


Figure 3: Cube Structure

Hierarchy Created

A time-based hierarchy was created in the Date dimension:

Year → Month Year → Date

This hierarchy enables drill-down analysis.

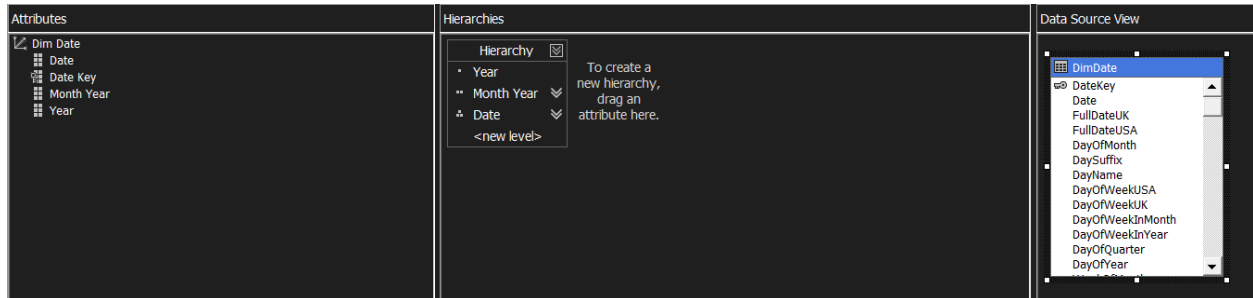


Figure 4: Date Hierarchy

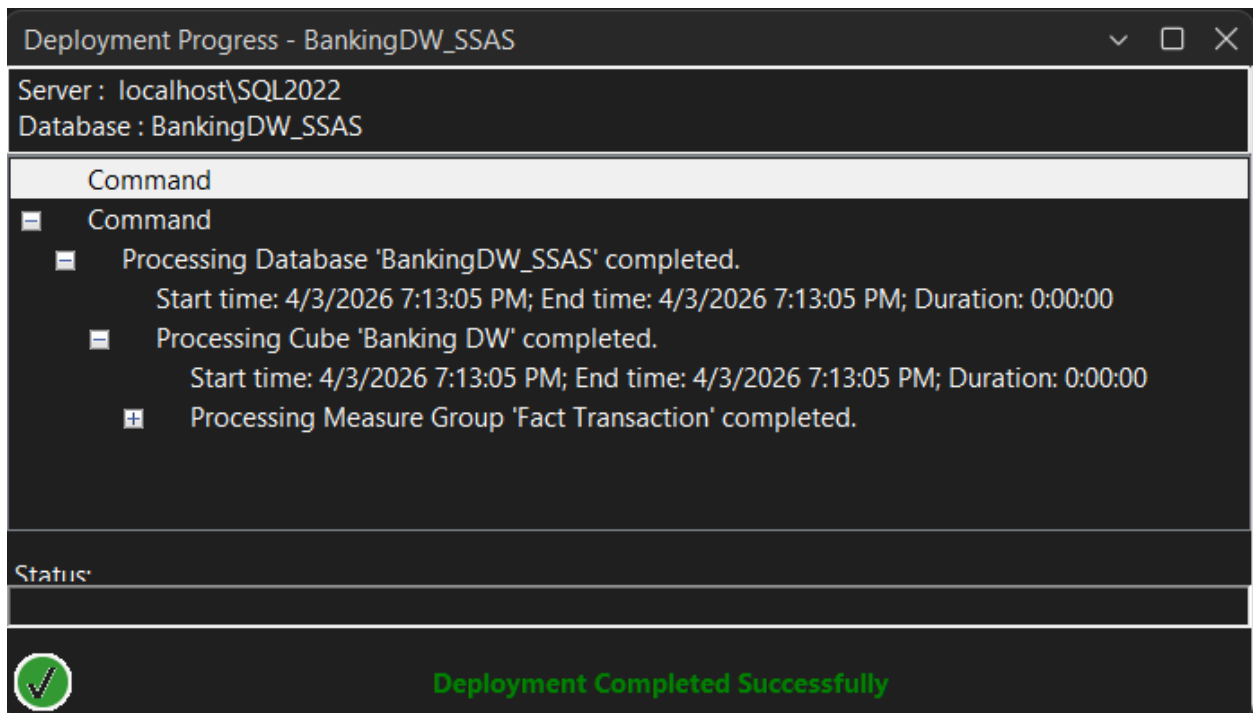


Figure 5: Deployment Success

4. OLAP OPERATIONS USING EXCEL

Excel was connected to the SSAS cube to perform OLAP analysis using Pivot Tables and Pivot Charts.

Roll-up

Aggregated transaction data by year.

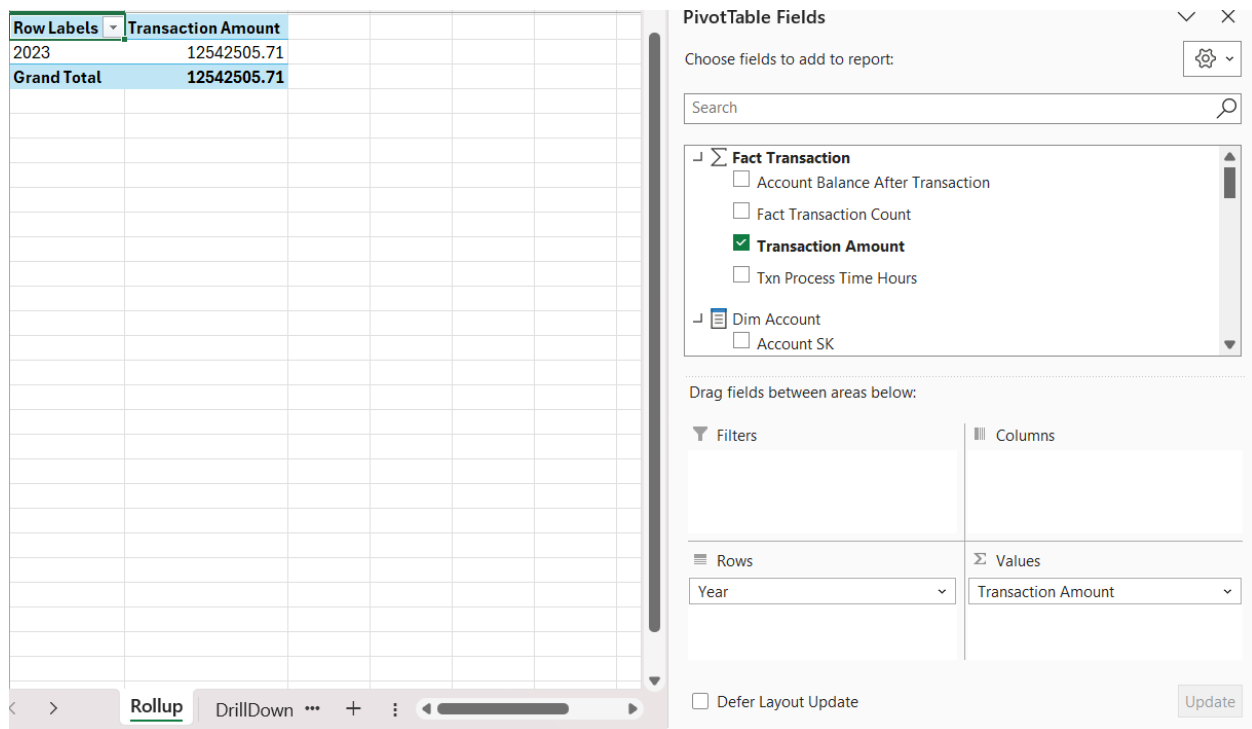


Figure 6: Roll-up

Drill-down

Expanded Year → Month analysis.

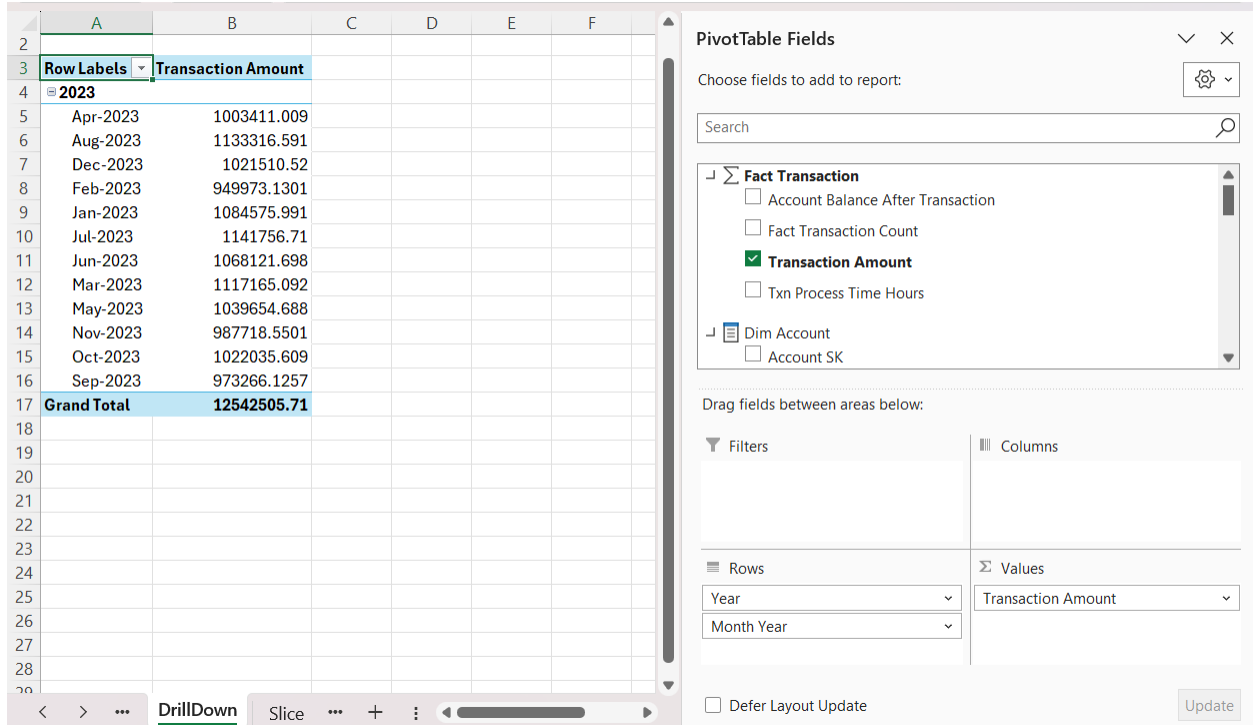


Figure 7: Drill-down

Slice

Filtered data by a specific dimension (e.g: Year).

The screenshot displays an Excel PivotTable and the PivotTable Fields task pane. The PivotTable is filtered by 'Month Year' to 'Apr-2023'. The 'Transaction Amount' is 1003411.009. The PivotTable Fields task pane shows 'Fact Transaction' with 'Transaction Amount' selected, and 'Dim Account' with 'Account SK' selected. The 'Rows' area shows 'Month Year' and the 'Values' area shows 'Transaction Amount'.

Row Labels	Transaction Amount
Apr-2023	1003411.009
Grand Total	1003411.009

PivotTable Fields

Choose fields to add to report:

Search

Fact Transaction

- ☐ Account Balance After Transaction
- ☐ Fact Transaction Count
- ☒ Transaction Amount
- ☐ Txn Process Time Hours

Dim Account

- ☐ Account SK

Drag fields between areas below:

Filters	Columns

Rows	Values
Month Year	Transaction Amount

☐ Defer Layout Update

Update

Figure 8: Slice

Dice

Applied multiple filters across dimensions. the data was filtered by City and Month Year to analyze a smaller subset of transactional data.

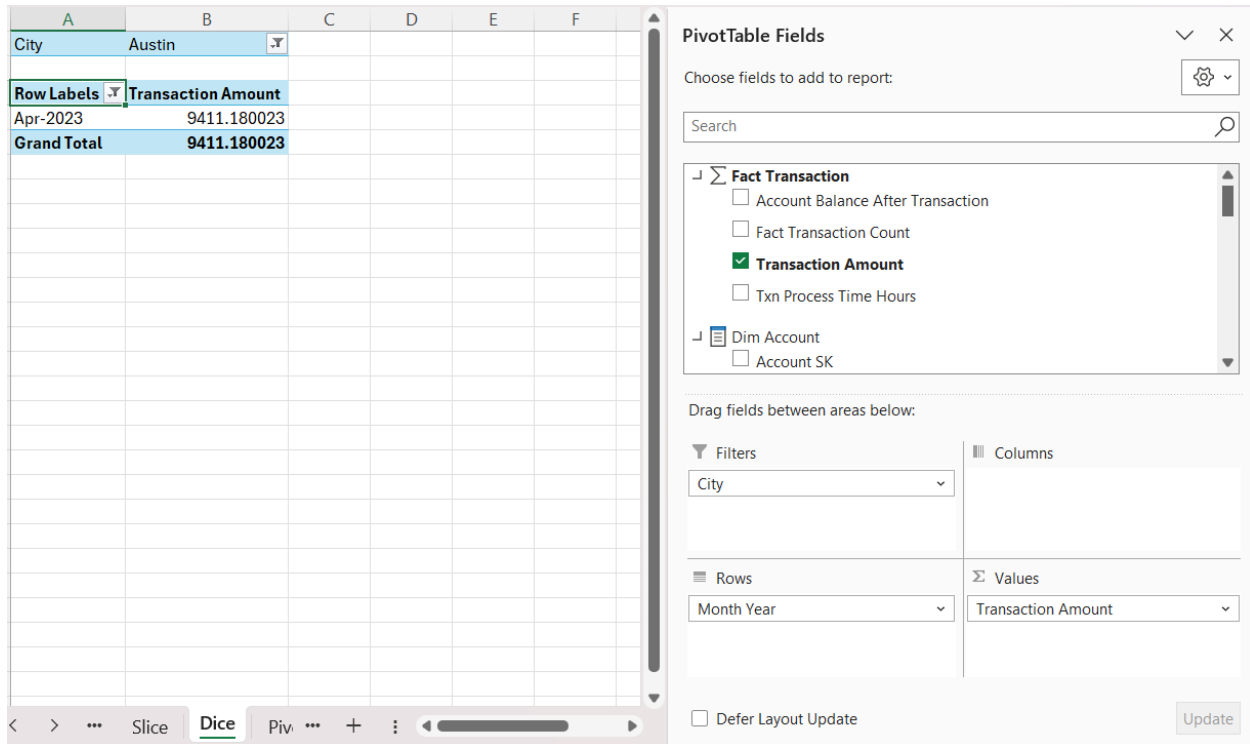


Figure 9: Dice

Pivot

Rearranged rows and columns.

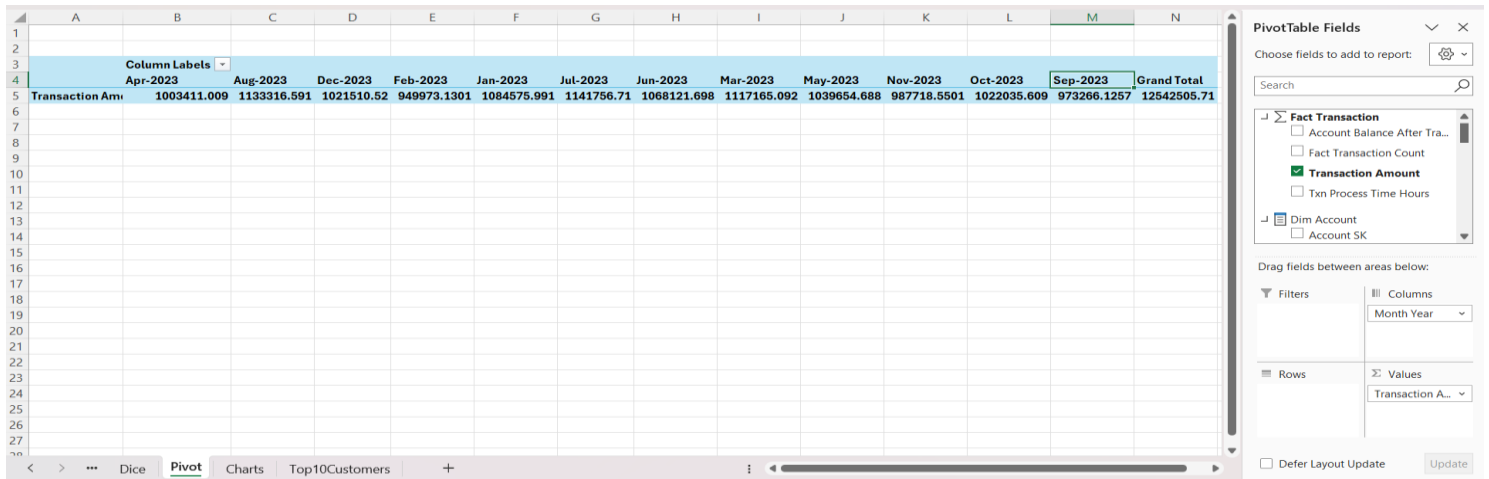
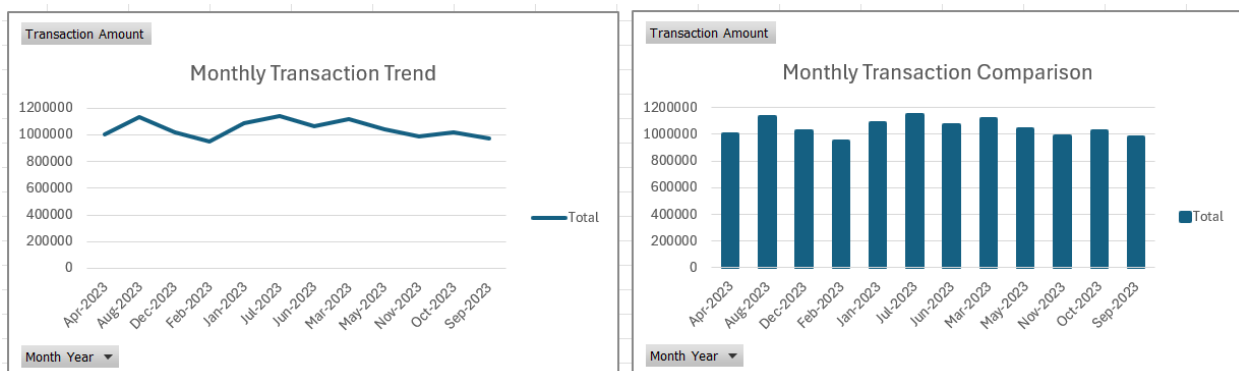


Figure 10: Pivot

Charts Created

- Line Chart - Monthly Transaction Trend
- Column Chart - Monthly Transaction Comparison
- Bar Chart - Top 10 Customers by Transaction Amount



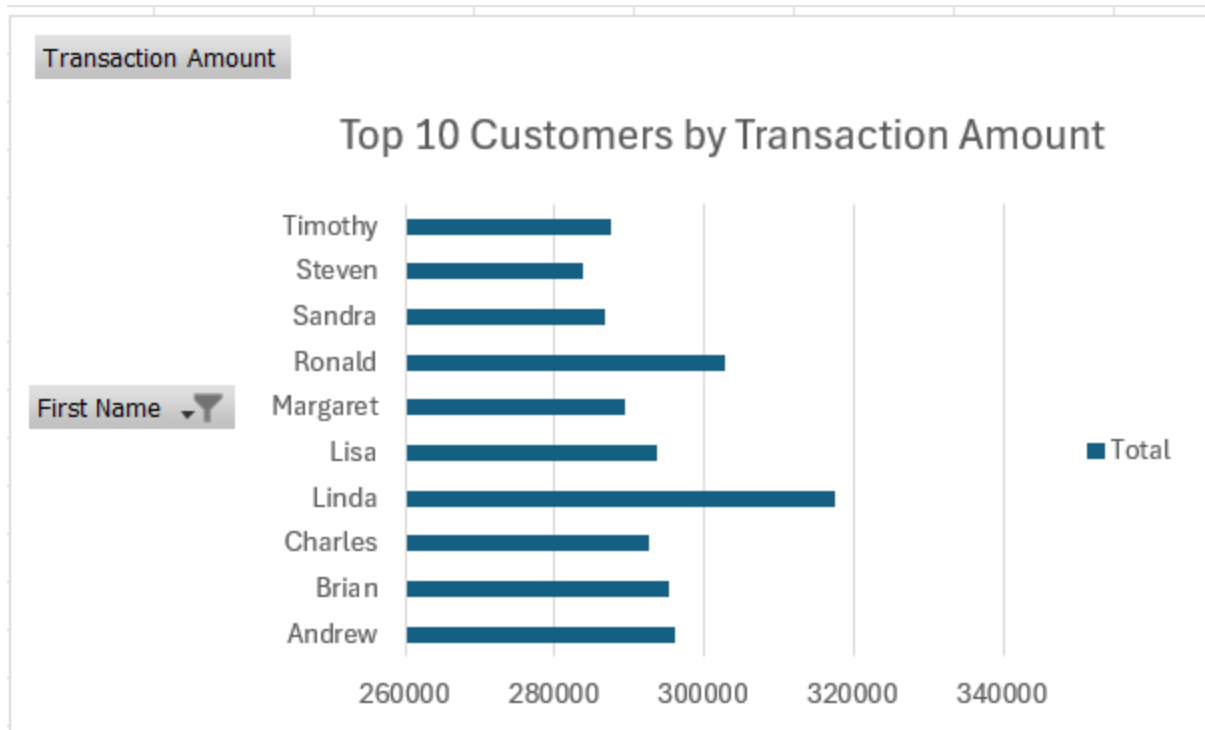


Figure 11: Charts

5. POWER BI REPORTS

Power BI was used to create interactive dashboards by connecting to the SSAS cube using a live connection.

Report 1 – Matrix Visual

A matrix was created to display grouped tabular data:

- Rows - Year, Month Year
- Columns - Account Type
- Values - Transaction Amount

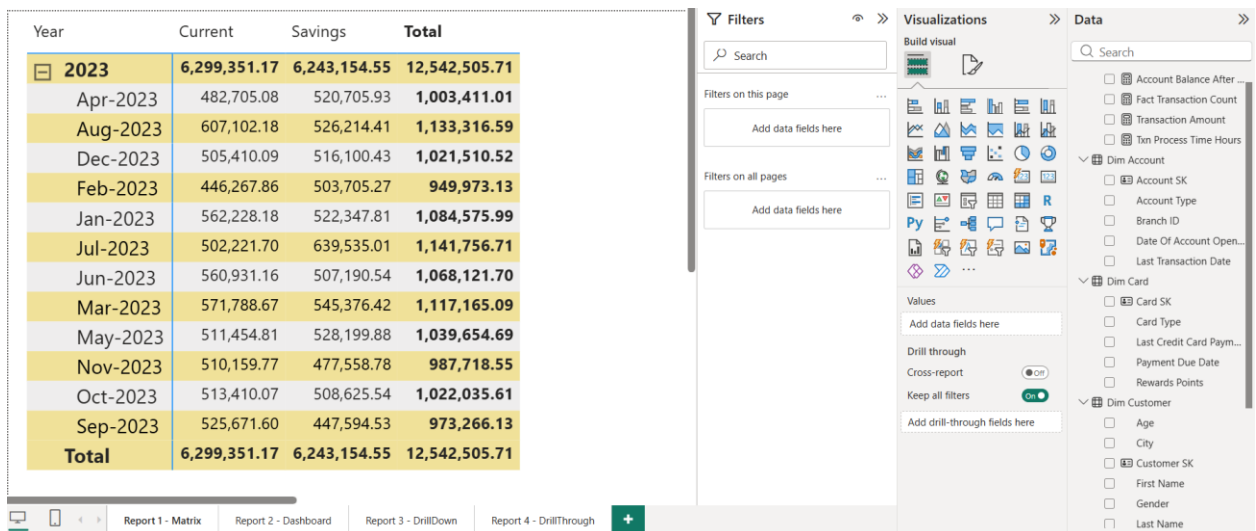


Figure 12: Matrix visual

Report 2 – Slicers with Dashboard

Multiple slicers were used:

- Slicer 1 → Year
- Slicer 2 → Month Year

Visuals included:

- Column Chart
- Line Chart
- Bar Chart

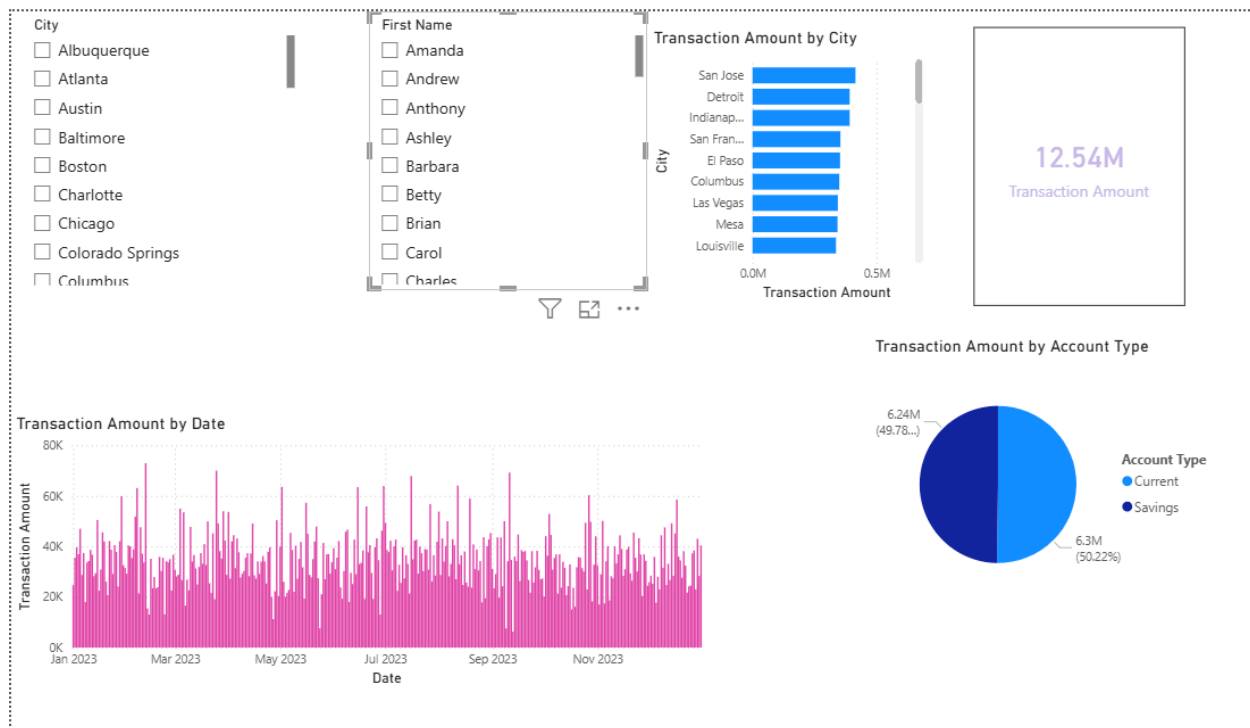


Figure 13: Full Dashboard/Before Filter

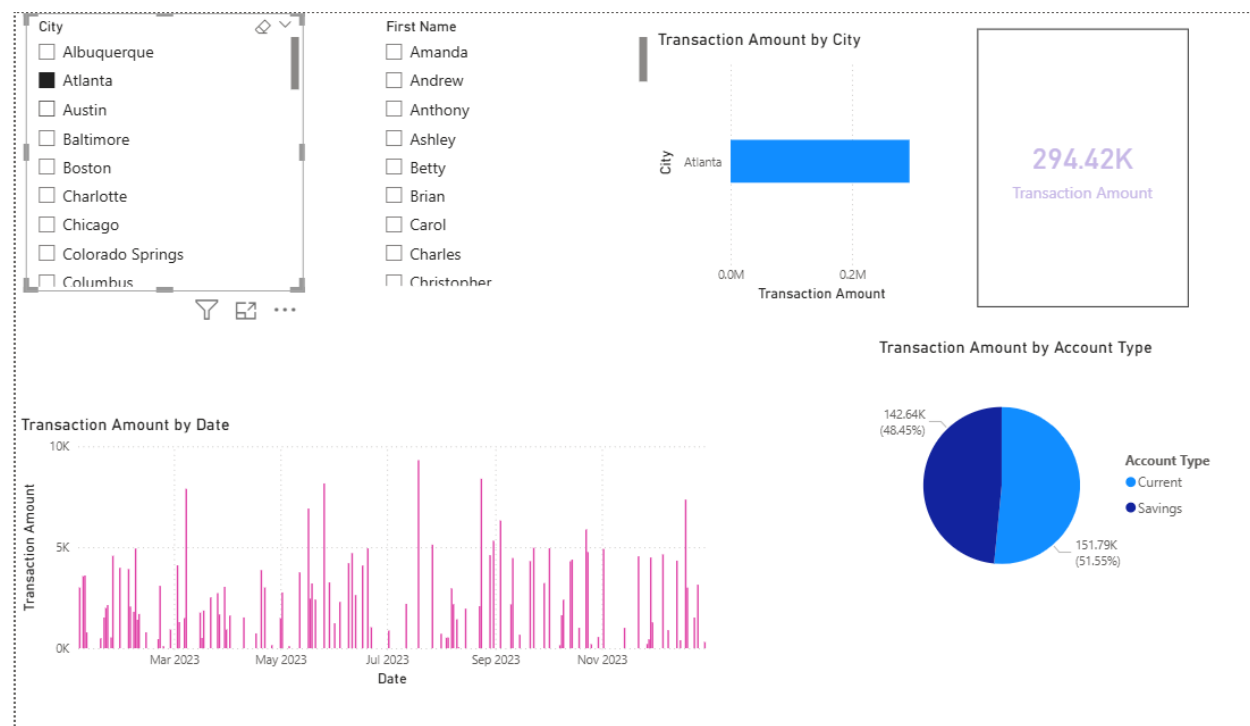


Figure 14: After Filter

Report 3 – Drill-down Report

A hierarchical drill-down was implemented using the date hierarchy:

Year → Month Year → Date

Users can expand data interactively.

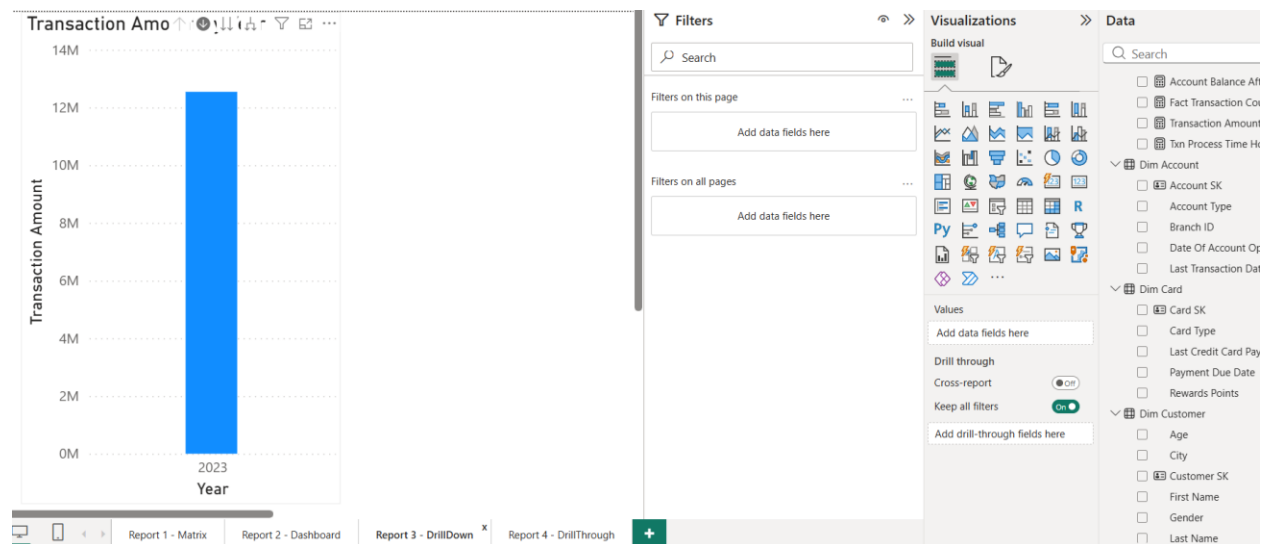


Figure 15: Drill down Report

Report 4 – Drill-through Report

Users can right-click on a visual and navigate to a detailed report page showing related data.

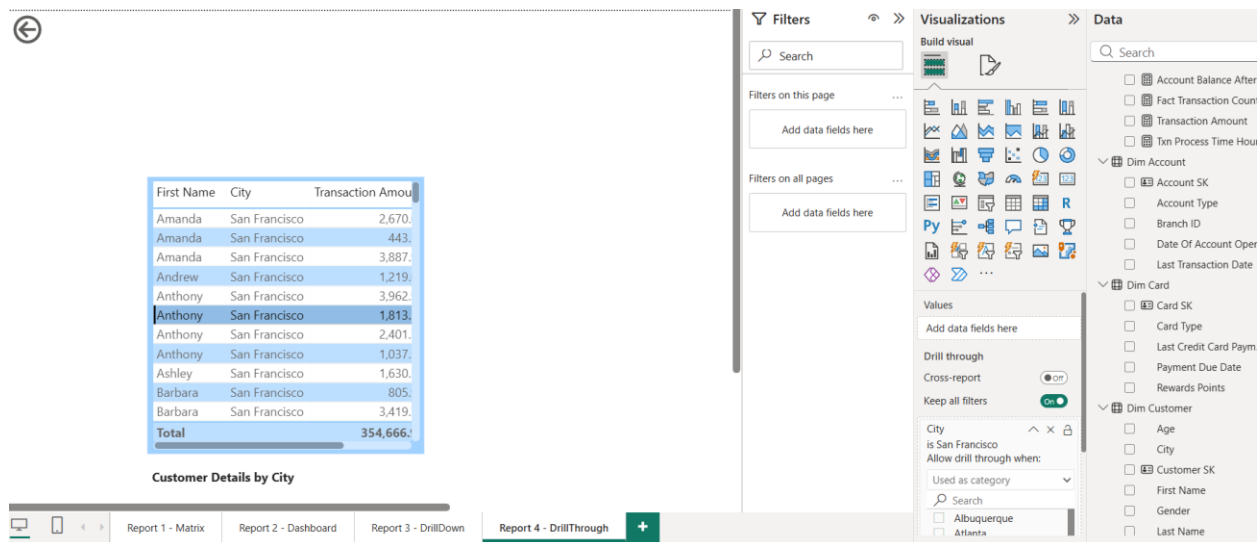


Figure 16: Drill-through Report

Data Connection

Power BI was connected using a live connection to the SSAS cube, ensuring real-time data analysis.

6. CONCLUSION

This project successfully illustrated a complete Business Intelligence solution. The SSAS cube allowed for multidimensional analysis; Excel would then perform the OLAP operations. Power BI dashboards enabled interactive and real-time visualizations for users to interact with their data seamlessly.

With these tools interconnected, data analysis capabilities were improved, paving the way for effective decision-making processes.